Multispectral Land-Cover Classification with EuroSAT: Does Spectral Range Beyond RGB Matter?

1. Problem Formulation

EuroSAT (Helber et al., 2019) is a land-cover benchmark of 27,000 Sentinel-2 patches, each 64 × 64 pixels with 13 spectral bands, labelled into ten classes: AnnualCrop, Forest, HerbaceousVegetation, Highway, Industrial, Pasture, PermanentCrop, Residential, River and SeaLake. The problem is single-label patch classification.

Sentinel-2 records bands the eye cannot see, in the near-infrared (NIR), red edge and short-wave infrared (SWIR). Do they buy better accuracy than a model that sees only red, green and blue? And if so, can I get most of that benefit more cheaply by handing spectral indices to an RGB model, rather than training on the full band stack?

2. Data Preprocessing

Loading. The patches are GeoTIFFs, read with rasterio. One quirk matters: EuroSAT stores B8A (narrow NIR) last, at file index 12, not at its canonical position 8. I reorder all 13 channels into the SSL4EO-S12 canonical order (Wang et al., 2023) before selecting any bands, which prevents the network from silently receiving mismatched channels.

Channel selection. Two views are taken from the same cube.

- *RGB:* canonical indices [3, 2, 1], i.e., B04 (red), B03 (green), B02 (blue).
- *MS:* canonical indices [1–8, 11, 12], giving B02–B08, B8A, B11 and B12 — ten bands. I drop B01, B09 and B10. Those three describe atmospheric absorption and aerosol scatter, not the surface, so in bottom-of-atmosphere data they hold no land-cover signal.

Preprocessing. Digital numbers are divided by 10,000 for approximate surface reflectance, then each patch is resized to 256 × 256 and centre-cropped to 224 × 224, the input size the backbone expects.

Augmentation. During training, I apply random horizontal and vertical flips and 90° rotations. At the patch scale, a field is a field whichever way it is turned, so every orientation is a valid view.

Split. I split the 27,000 patches once using stratified sampling: 21,600 (80%) for training and 5,400 (20%) held out for testing, and saved to data/splits.npz. A validation set (15% of the training pool, 3,240 patches) is carved out at runtime and never written to disk, so the test set stays untouched across every experiment.

3. Features / Model Architecture

3.1 CNN backbone

Both modalities use ResNet-18 from TorchGeo (Stewart et al., 2022) with Sentinel-2 MoCo weights: SENTINEL2_RGB_MOCO (3 channels) for RGB, SENTINEL2_ALL_MOCO (13 channels) for MS.

The 13-channel first convolution (weights of shape 64 × 13 × 7 × 7) will not take a 10-channel input. I replace it with a Conv2d(10, 64, 7, 7) initialised by slicing the pretrained weights to the ten kept bands: new_conv1.weight = pretrained_conv1.weight[:, KEPT_MS_IDX, :, :]. The slices for the three atmospheric bands are discarded; the ten land-band slices are kept as pretrained, so each kept band still meets its own filter.

The classification head is a small two-layer MLP, Linear(512 → 256) → ReLU → Linear(256 → 10). I fine-tune with separate learning rates for the backbone (1×10^{-4}) and head (1×10^{-3}), a batch size of 64, mixed precision, and early stopping on validation accuracy (patience 10). Training ran on a single RTX 4060 Laptop GPU (8 GB) under PyTorch 2.3.1 / CUDA 12.1, at roughly 95–100 s per epoch.

3.2 Spectral index features (32-d)

Eight physics-based indices are computed from the raw patch reflectance:

Index	Formula	Sensitivity
NDVI	(B08−B04)/(B08+B04)	Vegetation density
NDRE	(B08−B05)/(B08+B05)	Crop type / red edge
NDWI	(B03−B08)/(B03+B08)	Open water
MNDWI	(B03−B11)/(B03+B11)	Water vs built-up
NDWI_Gao	(B08−B11)/(B08+B11)	Vegetation water content
NDBI	(B11−B08)/(B11+B08)	Impervious surfaces
BSI	((B11+B04)−(B08+B02))/…	Bare soil
SAVI	1.5×(B08−B04)/(B08+B04+0.5)	Soil-adjusted vegetation

For each index I take four statistics over the patch — mean, std, p10, p90 — giving 8 × 4 = 32 features per sample. No atmospheric bands are used.

3.3 LightGBM fusion arms

I take the 512-d global-average-pooled embedding from each fine-tuned backbone (head removed) and the spectral-index vectors, and combine them into five feature sets for a LightGBM classifier, with early stopping on the carved validation set.

4. Experiments Performed

ID	Arm	Input features	Dim
E1	indices_only	Spectral indices only	32
E2	rgb_cnn	Fine-tuned RGB CNN	512
E3	rgb_cnn_indices	RGB CNN + spectral indices	544
E4	ms_cnn	Fine-tuned MS CNN	512
E5	ms_cnn_indices	MS CNN + spectral indices	544

E3 is the experiment I care about most: can feeding NIR/SWIR information to an RGB model through indices close the gap to a model trained end-to-end on the bands? E5 asks the opposite — once the CNN already sees all ten bands, do the indices still add anything? The two end-to-end CNNs give the straight RGB-versus-MS comparison.

5. Evaluation Results

CNN end-to-end (test set, n = 5,400)

Model	Test Acc	Macro-F1	Epochs	Best Val Acc
RGB CNN	97.80%	97.76%	23	98.55%
MS CNN	98.65%	98.62%	19	99.07%

LightGBM arms (test set, n = 5,400)

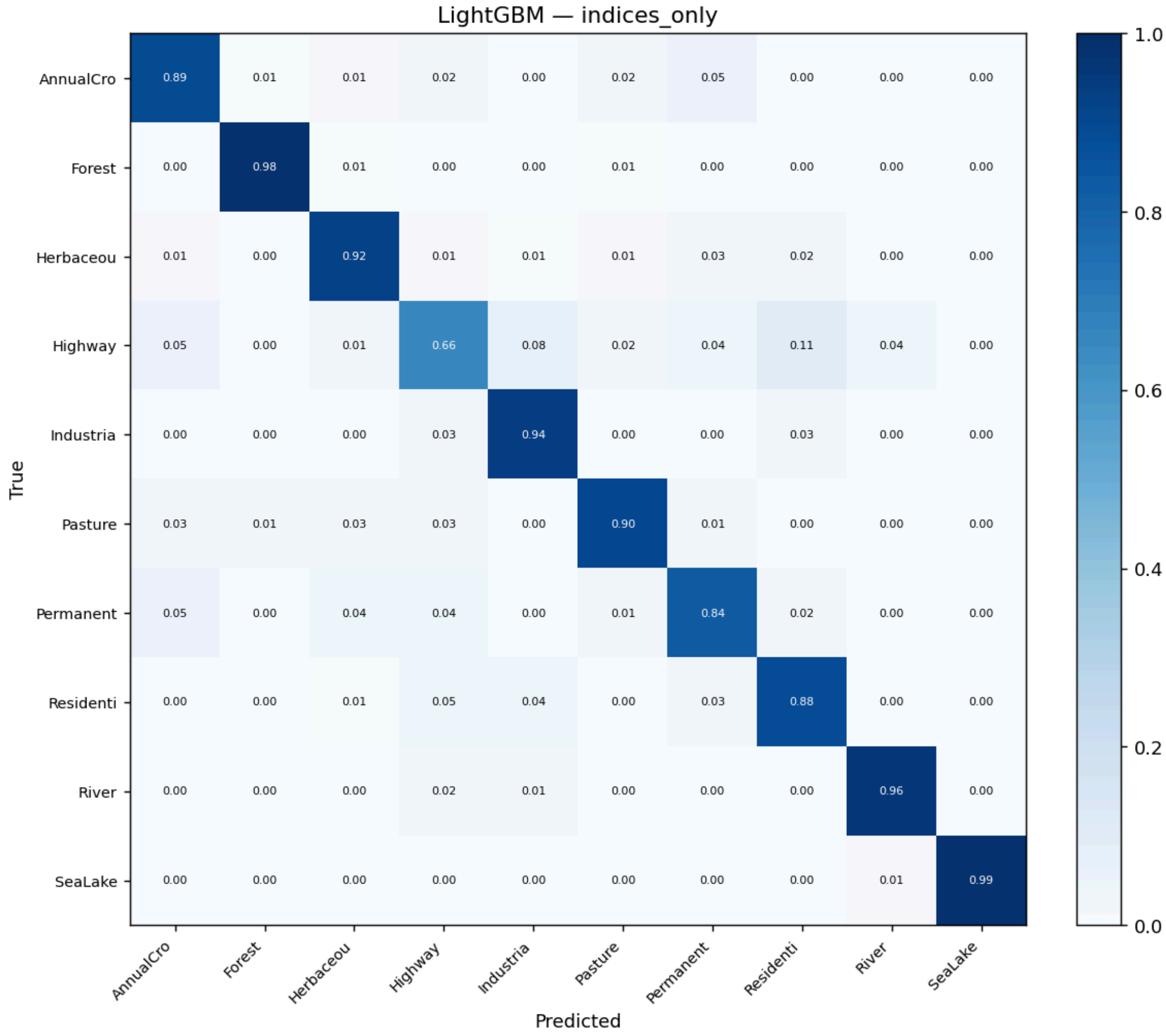

Figure 1. Indices-only confusion matrix.

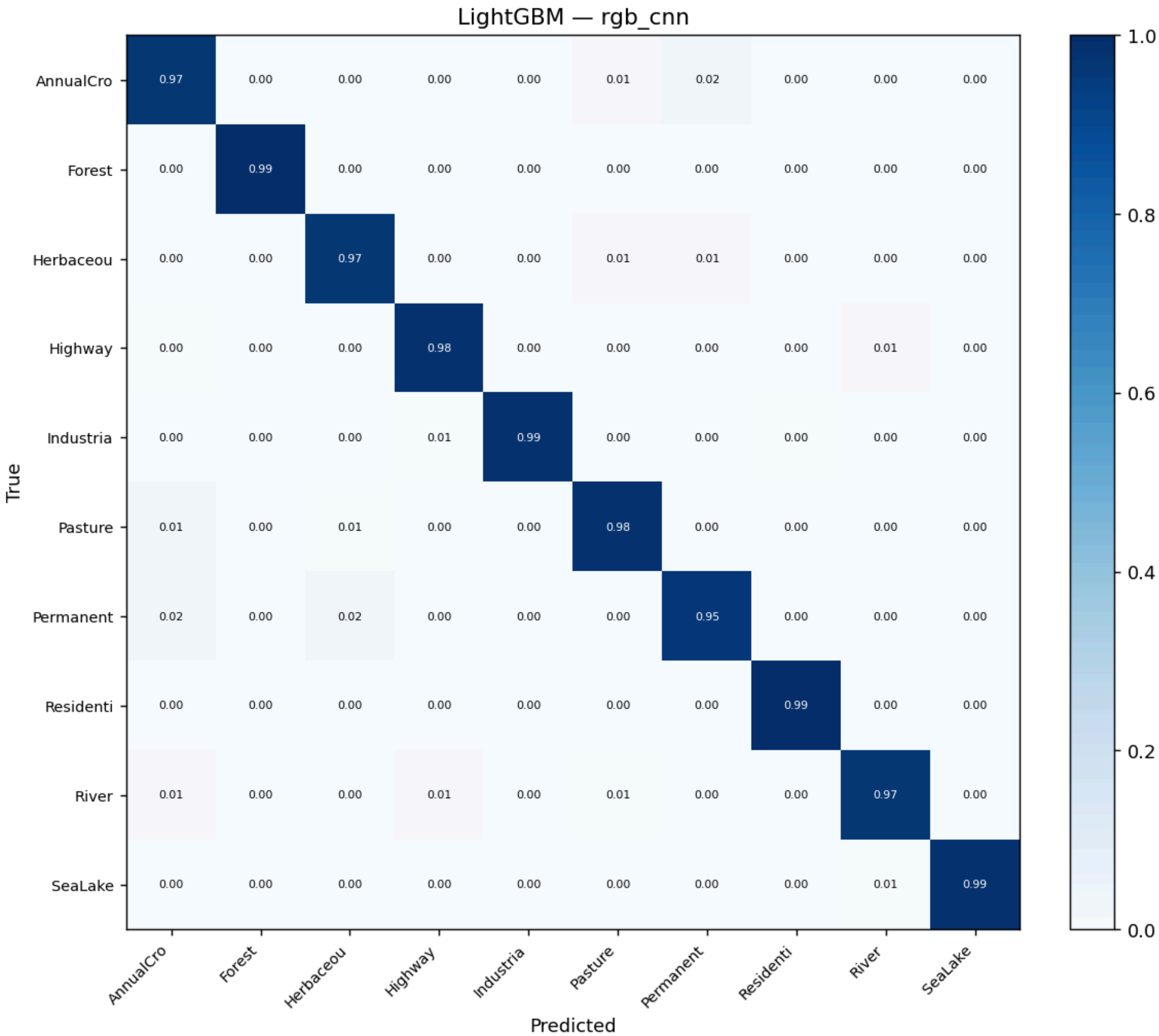

Figure 2. RGB CNN confusion matrix.

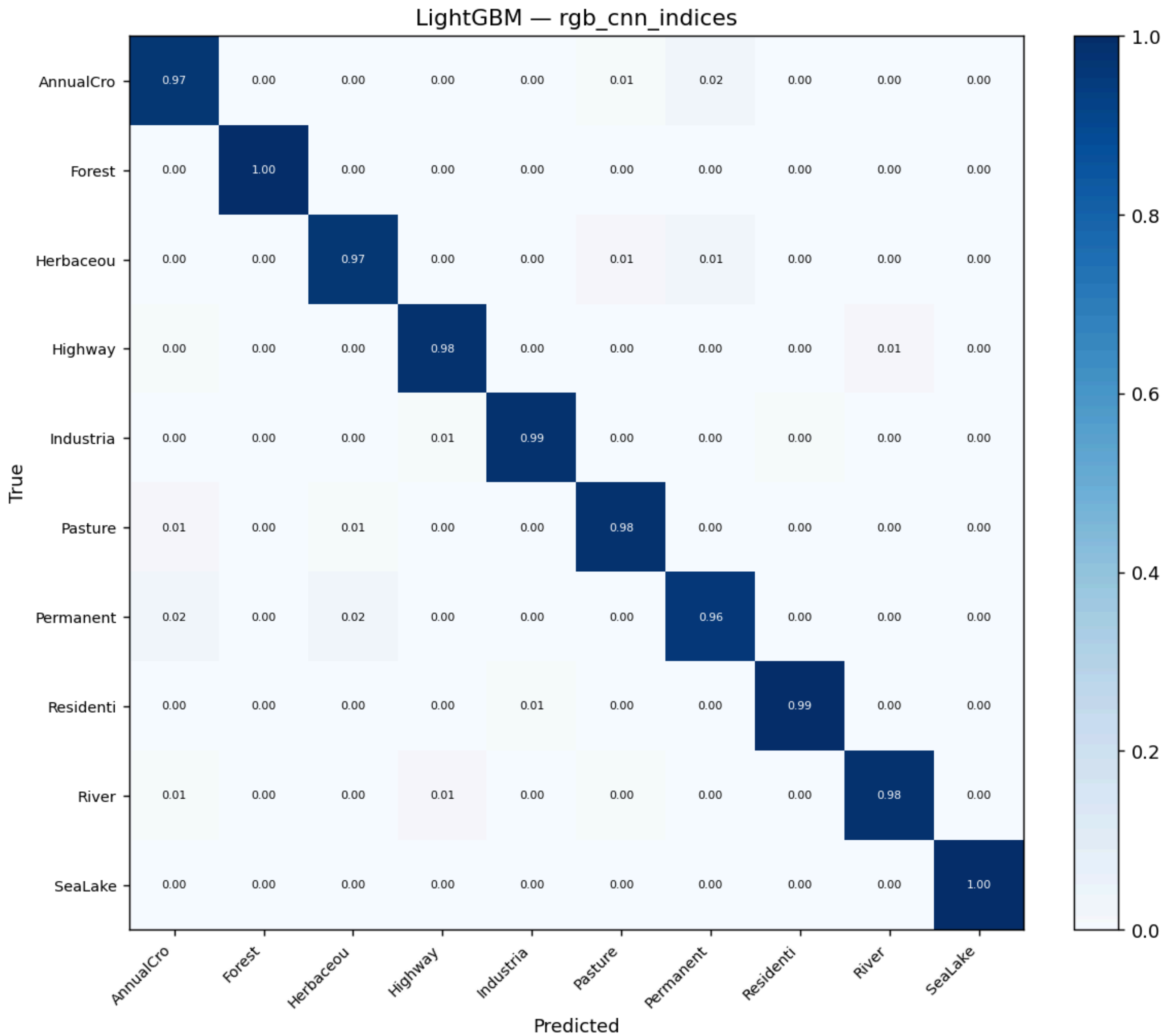

Figure 3. RGB CNN + indices confusion matrix.

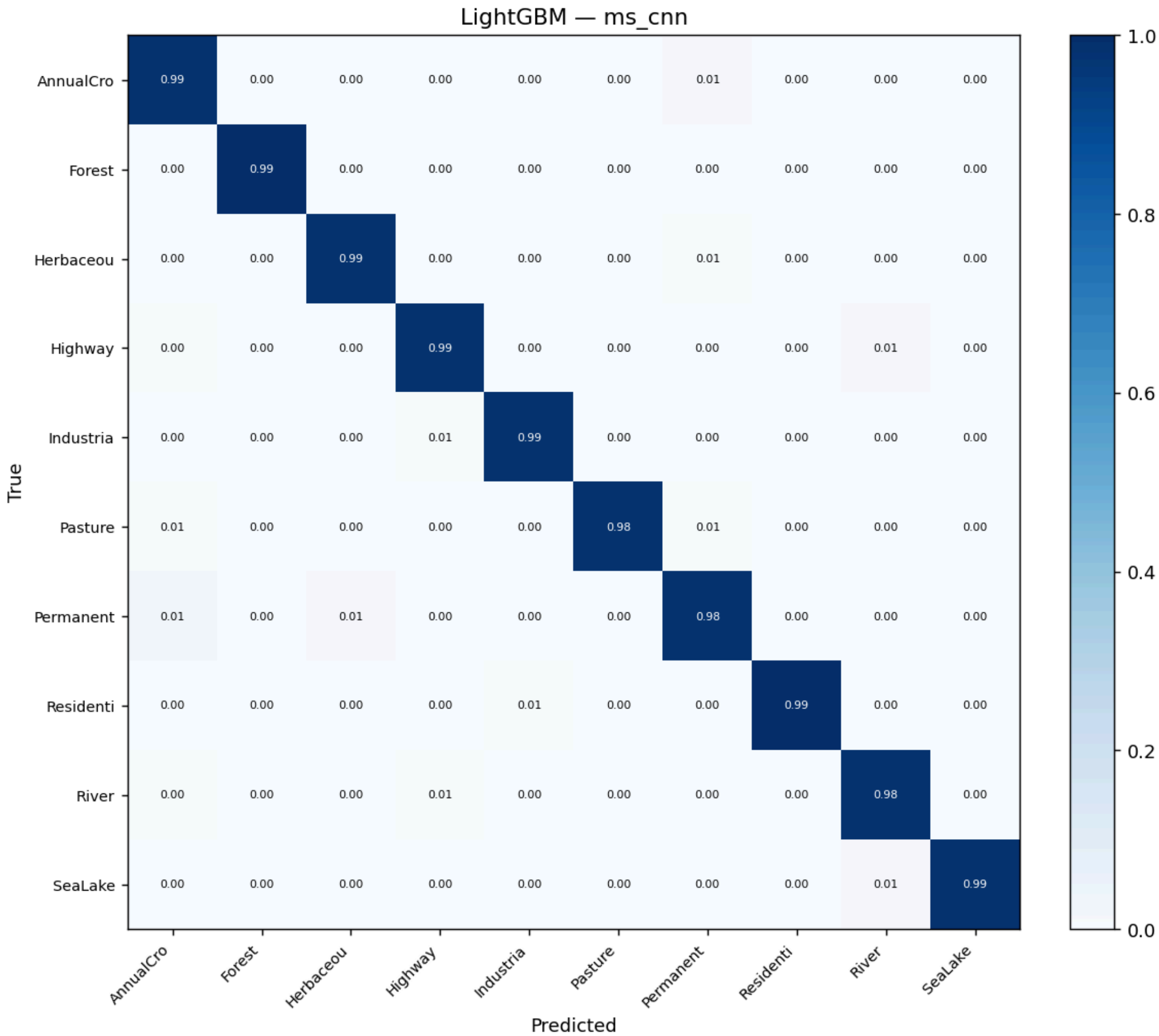

Figure 4. MS CNN confusion matrix.

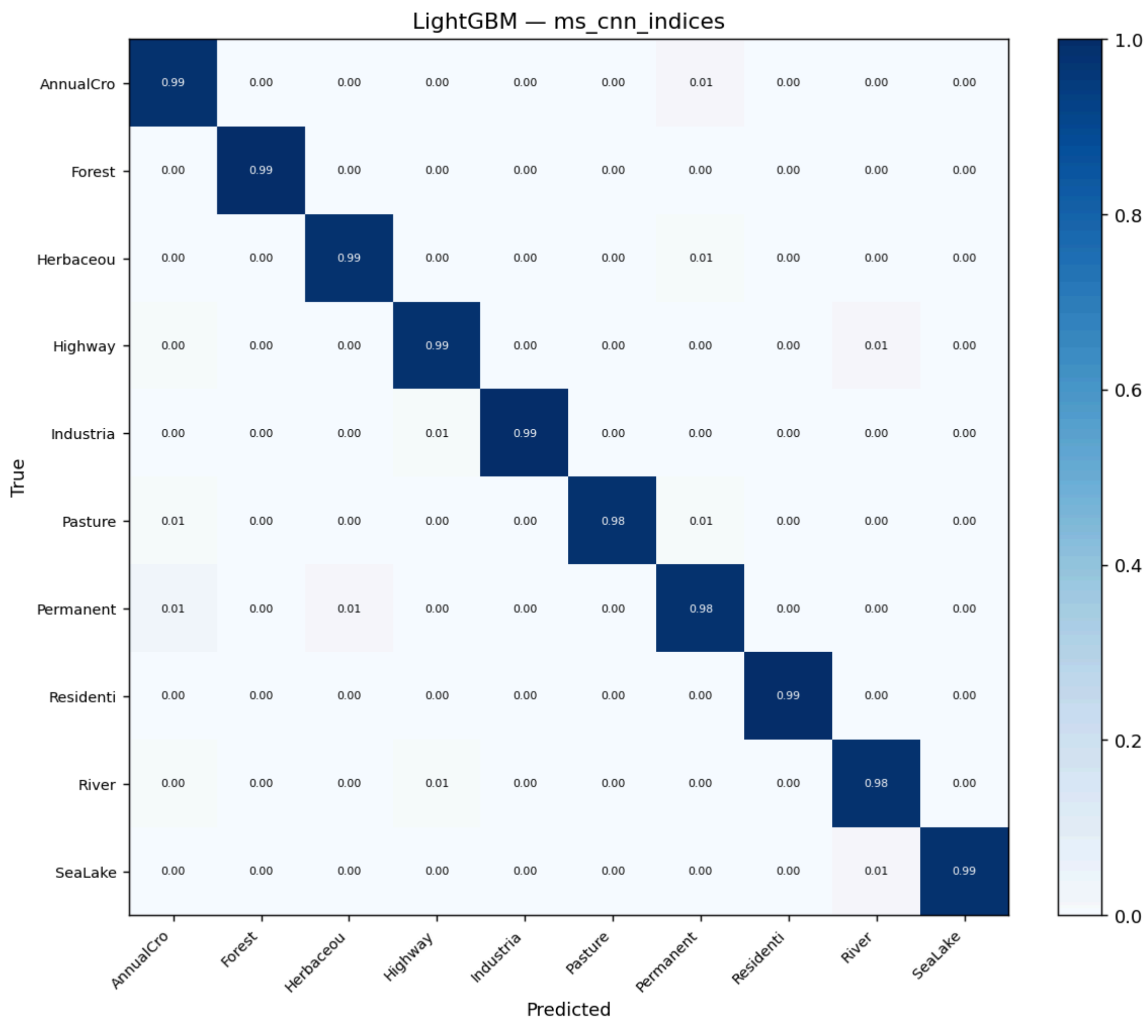

Figure 5. MS CNN + indices confusion matrix.

Arm	Dim	Test Acc	Macro-F1
indices_only	32	89.87%	89.44%
rgb_cnn	512	98.02%	97.96%
rgb_cnn_indices	544	98.19%	98.13%
ms_cnn	512	98.85%	98.83%
ms_cnn_indices	544	98.85%	98.83%

McNemar significance tests (n = 5,400)

These tests compare paired predictions of the LightGBM arms, not the end-to-end CNNs.

Comparison	χ^2	p	Significant
rgb_cnn vs ms_cnn	18.09	2.1×10^{-5}	Yes
rgb_cnn_indices vs ms_cnn	12.01	5.3×10^{-4}	Yes
rgb_cnn vs rgb_cnn_indices	3.37	0.066	No
rgb_cnn_indices vs ms_cnn_indices	11.78	6.0×10^{-4}	Yes

Per-class highlights (F1)

The MS model's gains over RGB land almost entirely on vegetation and crop classes: PermanentCrop rises from 0.948 to 0.976, HerbaceousVegetation from 0.963 to 0.981, AnnualCrop from 0.968 to 0.980. SeaLake and Forest are already at the ceiling in both models (F1 ≥ 0.986), so there is little room to move. The indices-only model, by contrast, falls apart on Highway (F1 0.706) and PermanentCrop (F1 0.834) — classes that hinge on spatial layout or on band contrasts four statistics cannot capture.

Confusion matrices, curves and galleries are under results/rgb/ and results/ms/; SHAP plots in results/ml/.

6. Key Results

1. **Multispectral input wins.** The MS CNN reaches 98.65% against the RGB CNN's 97.80%, a 0.85-point gain, and at the feature level the MS embedding beats the RGB embedding by a statistically significant margin (ms_cnn vs rgb_cnn, McNemar $p = 2.1 \times 10^{-5}$).
2. **Bolting indices onto RGB does not really close the gap.** The +0.17-point gain from rgb_cnn to rgb_cnn_indices is not significant ($p = 0.066$), and rgb_cnn_indices still trails ms_cnn by a significant margin ($p = 5.3 \times 10^{-4}$). Hand-crafted summary statistics are not a full replacement for training on the bands.
3. **For the MS model, the indices are redundant.** ms_cnn_indices lands on the same 98.85% as ms_cnn; whatever the indices encode, the network has already learned it.

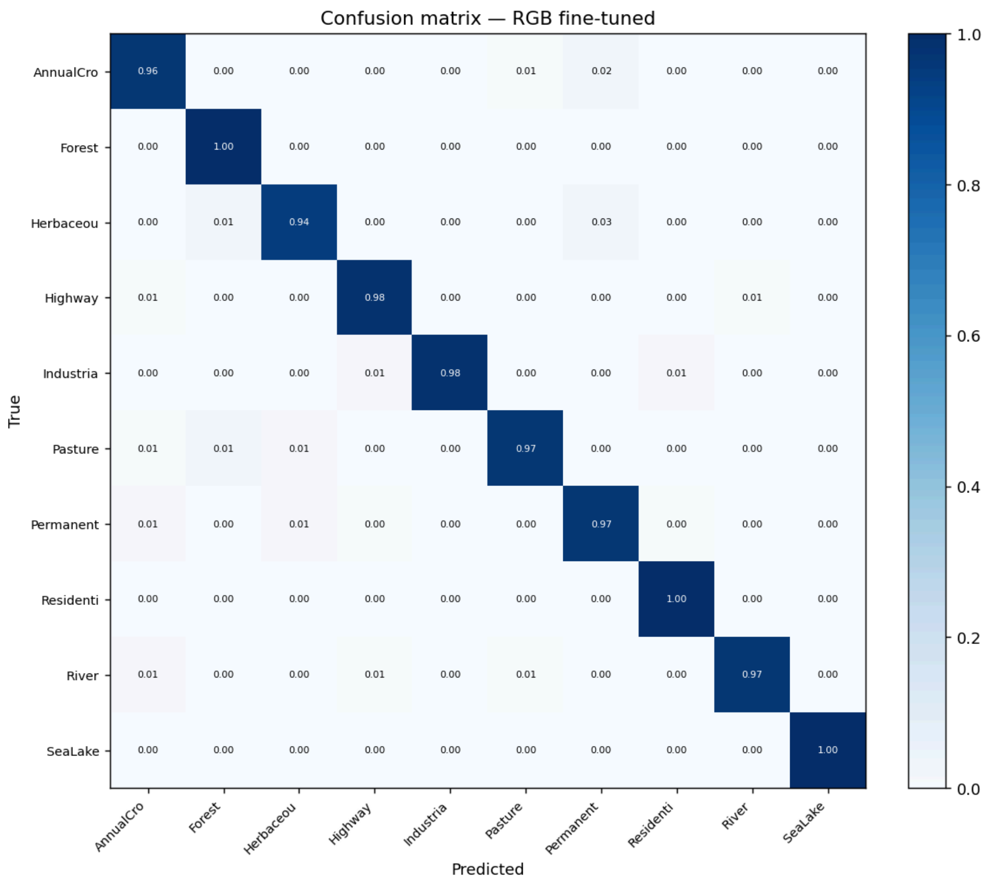

Figure 6. RGB CNN end-to-end confusion matrix.

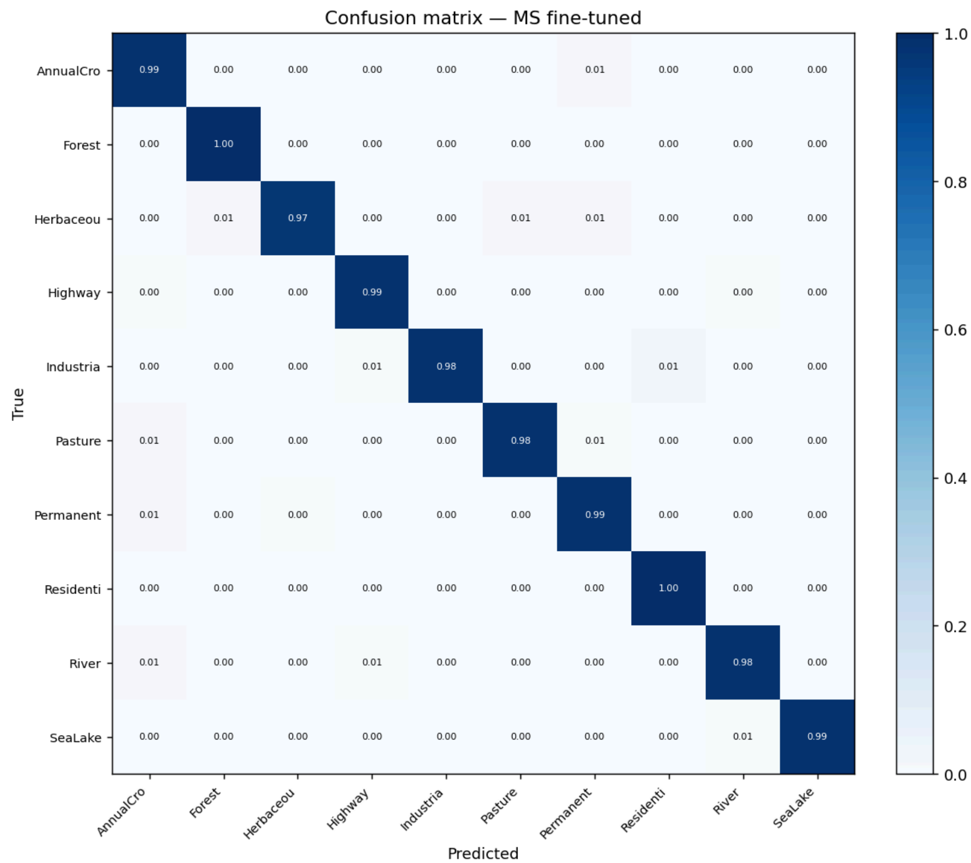

Figure 7 — MS CNN end-to-end confusion matrix.

7. Discussion of Findings

The MS advantage is small in absolute terms, 0.85 points end-to-end, as expected on a benchmark this saturated. What matters is where it shows up. PermanentCrop, HerbaceousVegetation and AnnualCrop are the classic RGB headache: different crops and grasslands look almost identical in visible light but pull apart sharply in B05–B08 (red edge and NIR), which track canopy structure and chlorophyll. Forest and SeaLake, already solved in RGB, barely move. The extra bands make their difference where RGB is genuinely ambiguous and have little effect where it is not.

E3 asks whether indices can paper over the RGB model's blindness in the infrared. They mostly cannot. The +0.17-point gain points the right way, since the indices do inject NIR/SWIR information the CNN never saw, but it does not reach significance ($p = 0.066$), and a gap to the MS model remains. Two things explain that residual. First, reducing each index to mean, std, p10, and p90 throws away all spatial arrangement: NDVI_p10 tells you the least-vegetated pixel in a patch but nothing about where it sits next to a road or a river, whereas the MS CNN learns those spatial relationships across all ten bands at once, and a 32-number vector simply cannot. Second, the SHAP analysis points to MNDWI_std and NDWI_std as the most important features overall, both measuring how mixed the water/built-up signal is within a patch. So the indices seem to earn their keep mainly at class boundaries and mixed pixels, which a convolution over multi-band feature maps already handles.

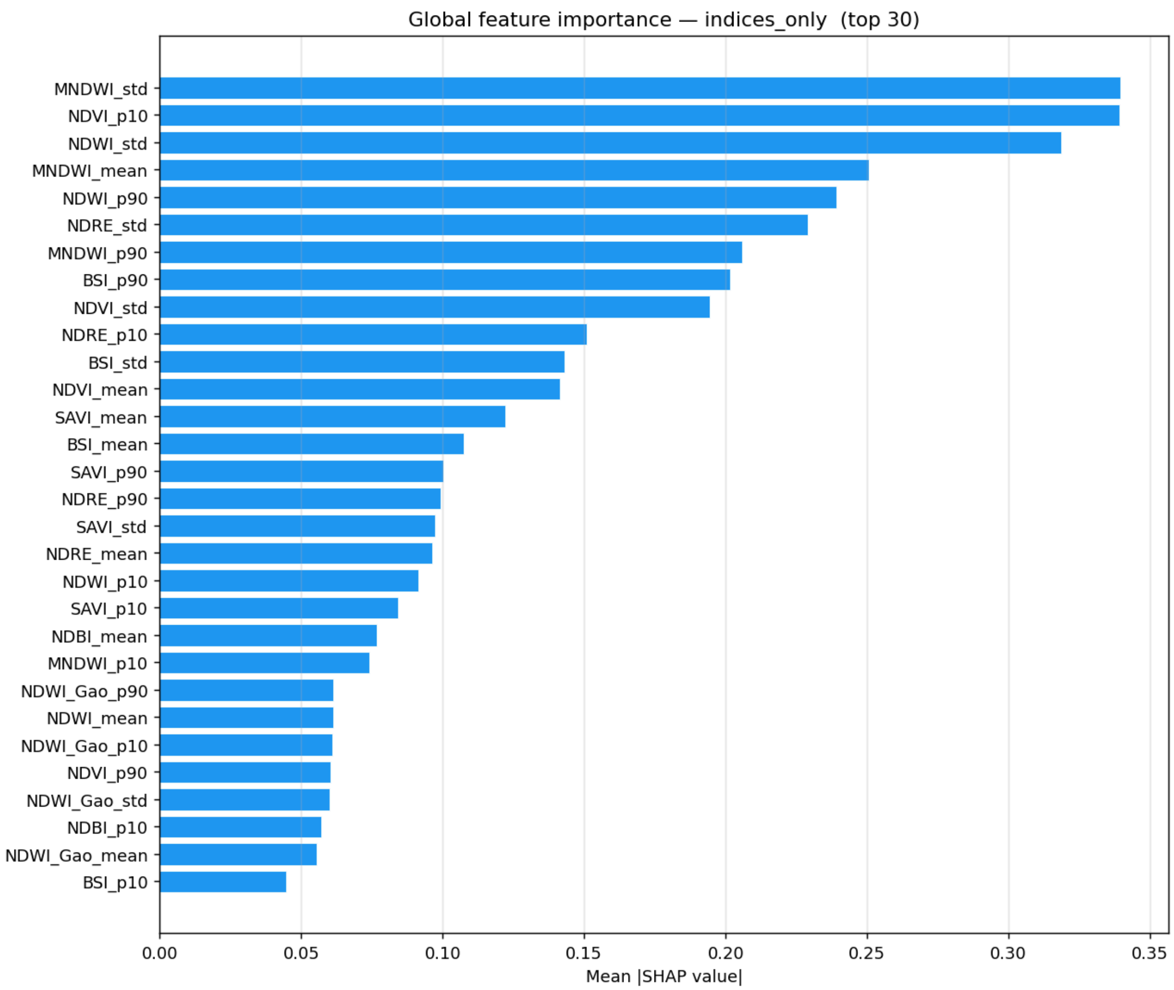

Figure 8. SHAP summary: MNDWI_std and NDWI_std dominate.

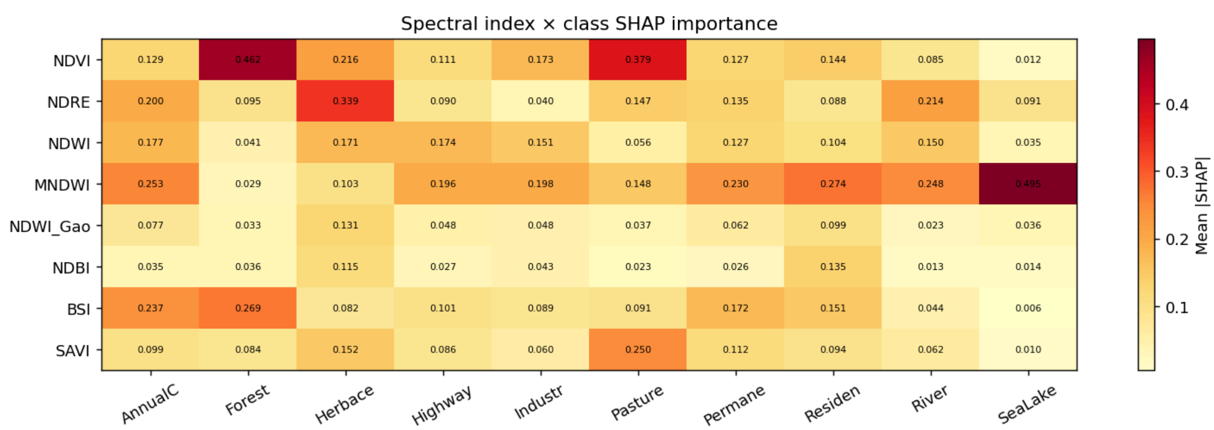

Figure 9. SHAP by class: NDVI and NDRE lead for crops.

E5 settles the converse. Adding the same indices to the MS model changes nothing — ms_cnn_indices and ms_cnn both sit at 98.85% accuracy and 98.83% macro-F1 — because the network, trained on ten bands, has already recovered whatever the ratios express. The contrast with E3 is the point: the indices only ever carry information the RGB feature space is missing, and even there the gain is not significant ($p = 0.066$), while the MS feature space already holds that information outright.

The gap between rgb_cnn_indices and ms_cnn_indices stays significant ($p = 6.0 \times 10^{-4}$), which says something practical: you cannot recover after the fact, by gluing on a few scalar summaries, what end-to-end training extracts from the bands. When NIR and SWIR are available at training time, train on them rather than approximate them with indices.

References

1. Helber, P., Bischke, B., Dengel, A., & Borth, D. (2019). EuroSAT: A novel dataset and deep learning benchmark for land use and land cover classification. IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing, 12(7), 2217–2226.
2. Stewart, A. J., Robinson, C., Corley, I. A., Ortiz, A., Lavista Ferres, J. M., & Banerjee, A. (2022). TorchGeo: Deep learning with geospatial data. Proceedings of the 30th International Conference on Advances in Geographic Information Systems (SIGSPATIAL).
3. Wang, Y., Braham, N. A. A., Xiong, Z., Liu, C., Albrecht, C. M., & Zhu, X. X. (2023). SSL4EO-S12: A large-scale multimodal, multitemporal dataset for self-supervised learning in Earth observation. IEEE Geoscience and Remote Sensing Letters, 20, 1–5.